NAVIN

Backend Developer

Accomplished IT professional with **13+ years** of experience in embedded systems design and cloud/backend development, along with proficiency in secure coding practices and recognized for developing multiple award-winning projects.

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https://github.com/navin-bhaskar

CORE COMPETENCIES

- Proven track record of designing and implementing robust applications tailored to client needs. Specialized in end-to-end application development, from conceptualization to deployment, leveraging cutting-edge technologies and best practices.
- Hands-on experience in implementing secure coding practices, and safeguarding systems against vulnerabilities. Skilled in developing IoT and AI solutions that integrate seamlessly with web and cloud infrastructures.
- Proficient in building scalable and robust back-end systems using modern frameworks and languages. In-depth working knowledge of NodeJS, React, FastAPI, Django, Flask, and Redis.
- Familiar with MongoDB, Git, MySQL, and Docker, ensuring efficient data management and deployment.
- Strong understanding of design patterns, object-oriented programming, and system architecture.
- Proven track record in leading development projects from concept to completion, consistently delivering solutions that enhance client satisfaction and operational efficiency.

WORK EXPERIENCE

Freelance AI engineer

Jan 2026 – Current *India*

- Built n8n automation workflows to power a RAG system, processing user-uploaded Google Drive screenshots using OCR to extract and index text for intelligent retrieval.
- Fine-tuned and trained multiple coding AI agents to enhance development productivity and accuracy.
- Architected and deployed a voice AI agent using ElevenLabs voice technology, integrated with a custom RAG pipeline driven by user-specific requirements.

Full Stack Staff Engineer

Incubyte

Jan 2025 –Nov 2025 *India*

- Agentic Tool Development (CrewAI): Designed and developed a multi-agent system using CrewAI to ingest medical provider expertise and generate human-readable summaries, demonstrating practical application of LLM-based workflows.
- Model Training & Optimization: Fine-tuned a local Ollama model to extract medical keywords from symptoms using synthetic data. This involved low-level model interaction and optimization for specific domain tasks (live demo available). (hosted here: https://navin-bhaskar-5--gradio-app-ui.modal.run/)
- LLM-Powered Automation: Engineered an agentic chat tool enabling customer support to execute complex, multi-step operations via natural language instructions, eliminating error-prone manual processes and reducing operational friction.

GENERAL SKILLS

Backend Development | Full Stack Development | Embedded Systems | System Programming | Web Development | IoT (Internet of Things) | JavaScript Development | Web Development | Data Structures | Microservices Architecture | Design Patterns | System Design | Project Management | Technical Troubleshooting | Stakeholder Management | Team Management

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java

Libraries and Frameworks: NodeJS, React, FastAPI, Django, Flask, Redis

Tools: MongoDB, Git, MySQL, Docker

Other Skills: Data structure and Algorithms, Microservices, Design Patterns, System Design

EDUCATION

Master of Science, Manipal Institute of Technology, India - 2012

CERTIFICATIONS

Certification: Specialized in Software Development & Problem Solving, Scaler - 2023

- ✓ Technical Leadership: Led a feature development team for a healthcare SaaS admin portal, focusing on system architecture and reliable integration of AI components.

Full Stack Engineer
Intel Corporation
Feb 2020 – Nov 2024 *India*

- ✓ Custom DSL Interpreter (Compiler Engineering): Designed and implemented a custom interpreter in Python using the LARK compiler toolkit for a domain-specific language. This replaced a legacy .NET solution, achieving a 25% increase in validation accuracy and a 50% reduction in processing time – demonstrating deep understanding of language parsing, AST manipulation, and execution semantics.
- ✓ Performance Optimization (Node.js Microservices): Developed API and executor microservices for a test suite validation system, delivering a 30% reduction in validation time through efficient code design and parallel execution patterns.
- ✓ System Tooling & Communication: Engineered SocketIO real-time communication and a Python CLI interface for client software, improving developer workflow and test automation reliability.

Firmware Engineer
Poynt, India *Apr 2018 – Dec 2019*

- ✓ Enhanced and developed firmware features for a payment system on the Cortex M3 platform, facilitating communication with an Android-based device.
- ✓ Created tools in Python for log analysis and debugging, significantly accelerating debugging processes and third-party firmware integration.

Firmware Engineer
NXP Semiconductors, India *Sep 2015 - Apr 2018*

- ✓ Added several features, such as context switching and user emulation, to a dual card architecture.
- ✓ Automated test case generation using Python for DESFire IP, reducing time to market by nearly 20%.

Firmware Engineer
Western Digital, India *Jun 2014 - Jul 2015*

- ✓ Implemented a "Design for Debug" feature enabling log collection on SanDisk's USB drives.
- ✓ Developed "peak power programming" to reduce power consumption during programming of 3D NAND.

Design Engineer
NXP Semiconductors, India *Oct 2012 – Jun 2014*

- ✓ Implemented target mode ISO 18092 (NFC target) emulation on LPC1768 MCU.
- ✓ Engineered a discovery loop for the NFC protocol on both initiator and target configurations.

Engineering Intern
NXP Semiconductors, India *Jun 2012 - Sep 2012*

- ✓ Developed multiple features for NXP's. NET-based "RFID Discover" application, improving PCSC reader functionality and logging, which resulted in a 30% increase in debugging efficiency.